A Matrix-Only, Self-Contained Proof of Tao's Theorem 1.3

"Almost all Collatz orbits attain almost bounded values"

Every step of the argument is presented as a matrix.

Legend: proved | structural reduction (proved) | irreducible analytic input

Matrix 0 — Objects and definitions (self-contained)

Object	Definition	Notes / Idris
Collatz map Col	$\mathrm{Col}(n) = \begin{cases} n/2 & n \text{ even} \\ 3n+1 & n \text{ odd} \end{cases}$	on $\mathbb{N}^+$; `Dynamics.Col`
Odd part odd	$\mathrm{odd}(n)$ = the odd number obtained by deleting all factors of 2 from n; $\mathrm{odd}(n) = n$ if n odd	$\mathrm{odd}(2^k q) = q$; `Dynamics.oddPart`, `oddFactor`
Syracuse map Syr	$\mathrm{Syr}(q) = \mathrm{odd}(3q+1)$, acting on odd q	$\mathrm{Syr} = \mathrm{odd} \circ \mathrm{Col}$ on odds; `Dynamics.Syr`
Iteration	$\mathrm{Col}^0 = \mathrm{id}$, $\mathrm{Col}^{k+1} = \mathrm{Col} \circ \mathrm{Col}^k$ (same for Syr)	`Core.iter`, `Core.iterPlus`
Orbit minimum	$\mathrm{Col}_{\min}(n) = \inf_{k \geq 0} \mathrm{Col}^k(n)$, $\mathrm{Syr}_{\min}(q) = \inf_{k \geq 0} \mathrm{Syr}^k(q)$	realised as "eventually below"
Eventually below	$\mathrm{Col}{\downarrow}(n, b) :\Leftrightarrow \exists k.\ \mathrm{Col}^k(n) \leq b$	`Core.EventuallyBelow`, `ColBelow`, `SyrBelow`
$f \to \infty$	$\forall t\, \exists T\, \forall x.\ x \geq T \Rightarrow f(x) \geq t$	`Large.TendsToInfinityOn`
Count	$\#\{\, m < N : p(m) \,\}$, written $\mathrm{cnt}_p(N)$	`Density.count`
Negligible (density 0)	p negligible $:\Leftrightarrow \forall k\, \exists N_0\, \forall N \geq N_0.\ (k+1)\, \mathrm{cnt}_p(N) \leq N$	`Density.Negligible`
Almost all	a.a. $p :\Leftrightarrow (\neg p)$ is negligible	`Density.AlmostAll`; `Large.AlmostAllOn`

The three headline statements are matrices of quantifiers:

$$\textbf{Thm 1.3 (Collatz)} : \forall f\, (f \to \infty) \;\Rightarrow\; \text{a.a. } n :\ \mathrm{Col}{\downarrow}\big(n,\, f(n)\big),$$

$$\textbf{Thm 1.6 (Syracuse)} : \forall g\, (g \to \infty) \;\Rightarrow\; \text{a.a. } q :\ \mathrm{Syr}{\downarrow}\big(q,\, g(q)\big),$$

$$\textbf{Thm 3.1 (quant.)} : \forall g\, (g \to \infty) \;\Rightarrow\; \{q :\ \neg\, \mathrm{Syr}{\downarrow}(q, g(q))\} \text{ has density } 0.$$

(`Large.Theorem13`, `Large.Theorem16`, `Dependencies.QuantitativeSyracuseBound`.)

Matrix 1 — The reduction chain (rows compose top to bottom)

#	From	To	Justification (key idea)	Idris
R1	Props. 1.9 & 7.8	Prop. 1.11 (stabilisation)	combine valuation tail estimate with renewal stability	`analyticInput-ToStabilisation`

#	From	To	Justification (key idea)	Idris
R2	Prop. 1.11	Thm 3.1	iterate the stabilised first-passage bound	`proposition11To-Theorem31`
R3	Thm 3.1	Thm 1.6	a density-0 exceptional set *is* the "almost all" conclusion	`theorem31To-Theorem16`
R4	Thm 1.6	Thm 1.3	odd-part orbit simulation + threshold compatibility (Matrix 2–3)	`theorem16To-Theorem13Constructive`
Σ	Props. 1.9 & 7.8	**Thm 1.3**	R4 $\circ$ R3 $\circ$ R2 $\circ$ R1	`structuredCentral-Theorem`

The composite is exactly the product of the four one-step reductions:

$$\underbrace{\text{Thm 1.3}}_{\text{Collatz}} \xleftarrow{\text{R4}} \underbrace{\text{Thm 1.6}}_{\text{Syracuse}} \xleftarrow{\text{R3}} \underbrace{\text{Thm 3.1}}_{\text{quantitative}} \xleftarrow{\text{R2}} \underbrace{\text{Prop 1.11}}_{\text{stabilisation}} \xleftarrow{\text{R1}} \underbrace{\text{Props 1.9, 7.8}}_{\text{analytic input}}.$$

Dependency (adjacency) matrix A, with $A_{ij} = 1$ iff row i is used to prove column j (reading "j depends on i"):

	1.9/7.8	1.11	3.1	1.6	1.3
1.9/7.8	0	1	0	0	0
1.11	0	0	1	0	0
3.1	0	0	0	1	0
1.6	0	0	0	0	1
1.3	0	0	0	0	0

(strictly upper triangular $\Rightarrow$ acyclic; a single directed path).

Matrix 2 — Step R4 proved: the odd-part orbit simulation

R4 is the only step with genuine dynamical content; it is fully proved. Its skeleton is the matrix of facts below (all in `OddPart`/`OddThreshold`).

#	Fact	Idris
a	n odd $\Rightarrow \mathrm{Col}(n) = 3n+1$; n even $\Rightarrow \mathrm{Col}(n) = n/2$	`colOddStep`, `colEvenStep`
b	Deleting factors of 2 terminates: $\mathrm{odd}(n)$ is odd for $n \geq 1$	`oddFactorOdd`, `syrIsOdd`
c	Iterating Col from n reaches $\mathrm{odd}(n)$ in `oddPartDropTime`(n) steps, with all intermediate heights $\geq \mathrm{odd}(n)$	`oddPartDropTimeNormalizesValue`
d	One Syr-step is realised by finitely many Col-steps: $\exists s.\ \mathrm{Col}^s(m) = \mathrm{Syr}^t(\mathrm{odd}(m))$ (as values)	`syrRealize`, `syrRealizeStep`
e	Hence a height-dominating semiconjugacy $\mathrm{odd} : (\mathbb{N}^+, \mathrm{Col}) \to (\mathrm{odds}, \mathrm{Syr})$: for all n, t $\exists s.\ \mathrm{Col}^s(n) \leq \mathrm{Syr}^t(\mathrm{odd}(n))$	`provenOddPartOrbitSimulation`
f	Simulations form a category (identity, composition)	`orbitSimulationId`, `orbitSimulationCompose`

Consequence (transfer). If $\mathrm{Syr}{\downarrow}(\mathrm{odd}(n), b)$ then $\mathrm{Col}{\downarrow}(n, b)$: pick t with $\mathrm{Syr}^t(\mathrm{odd}(n)) \leq b$; by row (e) some $\mathrm{Col}^s(n) \leq \mathrm{Syr}^t(\mathrm{odd}(n)) \leq b$. `oddPartOrbitTransfer`

Matrix 3 — Step R4 proved: the threshold is *constructed*

To run Thm 1.6 with a Syracuse growth function whose value at $\mathrm{odd}(n)$ never exceeds $f(n)$, the threshold is *built* from the growth witness w of f (no separate hypothesis). Define, for odd q,

$$t_w(q) \;=\; \max\{\, t \le q : \ \texttt{thresholdFor}(w)(t) \le q \,\}.$$

#	Property	Idris
g	$\mathrm{odd}(n) \le n$ (deleting factors of 2 cannot increase)	`oddFactorLe`, `oddPartSizeLe`
h	Compatibility: $t_w(\mathrm{odd}(n)) \le f(n)$	`oddThresholdOfCompatible`
i	Growth: $f \to \infty \Rightarrow t_w \to \infty$	`oddThresholdOfGrows`
j	Bounded search is correct (largest admissible target)	`findBest`, `findBestCompat`, `findBestGe`

Proof of R4. Given $g := t_w$, Thm 1.6 yields a.a. q : $\mathrm{Syr}{\downarrow}(q, t_w(q))$. Pull back along odd (Matrix 4, closure) and monotonicity in the bound (row h) gives a.a. n : $\mathrm{Syr}{\downarrow}(\mathrm{odd}(n), f(n))$; Matrix 2 (transfer) converts each to $\mathrm{Col}{\downarrow}(n, f(n))$. ☐ `liftSyracuseToCollatz`

Matrix 4 — The "almost all" algebra (density model, proved)

"Almost all" means the exceptional set has natural density 0. This is a genuine, non-degenerate measure of smallness; the closure laws it must obey are all proved from first principles.

Law	Statement	Idris
Monotone (subset)	$S \subseteq T$, S negligible only via T; a.a. closed under supersets	`negligibleMono`, `almostAllMono`
Finite ∪ / ∩	union of negligibles is negligible; intersection of a.a. is a.a.	`orNegligible`, `andAlmostAll`, `orListNegligible`, `andListAlmostAll`
Bounded ⇒ small	finite / cofinite sets are negligible / a.a.	`boundedNegligible`, `almostAllCofinite`
Complementary count	$\mathrm{cnt}_p(N) + \mathrm{cnt}_{\neg p}(N) = N$	`countComplement`
Full space	$\mathrm{cnt}_\top(N) = N$; $\mathrm{density}(\top) = 1$	`countAllTrue`
Non-degeneracy	the whole space is **not** negligible (else $2N \le N$)	`allNotNegligible`
Cofinal / teeth	a negligible set omits arbitrarily large n	`negligibleCofalse`, `almostAllCofinal`
On the carriers	same algebra on $\mathbb{N}^+$ (`Pos`) and odds (`OddPos`)	`CarrierDensity.*`

Pull-back and push-forward used by R3, R4 are the structural maps `controlPullback`, `controlMap` (`Large`); combined with the laws above they move "almost all" along odd and along pointwise implications.

Matrix 5 — What is proved vs. what is assumed

Ingredient	Status	Remark
Collatz/Syracuse dynamics	proved	computable; $\mathrm{Col}(3) = 10$, $\mathrm{Syr}(7) = 11$

Ingredient	Status	Remark
Odd-part correspondence (R4)	proved	Matrices 2–3, no `believe_me`
Threshold system	proved	*constructed* from the growth witness
Reduction plumbing R1–R3	proved	structural; transports its input verbatim
“Almost all” algebra	proved	density model, non-degenerate (Matrix 4)
Prop. 1.9 (valuation tail)	assumed	2-adic valuation distribution & sub-Gaussian tail
Prop. 7.8 (renewal stability)	assumed	monotonicity of the renewal process
Thm 1.3	**proved from the two inputs**	`structuredCentralTheorem`; provably equal to `centralTheoremUnconditional`

The single remaining mathematical dependency is the pair of deep analytic estimates (Props. 1.9 and 7.8), highlighted in red. Everything else — the entire logical skeleton, the odd-part dynamics, the constructed threshold, and the non-degenerate density model of “almost all” — is a total, machine-checked Idris proof containing no `believe_me` and no axioms.